

MacroWatch

Development Phases — Phase 1, Phase 2 & Phase 3

Version: v1.1-stable | Date: May 8, 2026

Overview

MacroWatch is a macro intelligence dashboard built around Ray Dalio's Big Debt Cycle framework. It aggregates real-time and near-real-time data from multiple sources to monitor US debt cycle dynamics, Treasury auction health, foreign capital flows, and key macro signals. Development is organized into three phases: Phase 1 (data foundation), Phase 2 (visualization and interpretation), and Phase 3 (investment action layer).

PHASE 1 — Data Foundation (COMPLETE)

Status: v1.1-stable tagged May 8, 2026

Phase 1 Objective

Build a fully functional dashboard with verified real data across all six tabs: Daily, Auctions, Weekly, Monthly, QTR, and Intel. Every data point sourced from official APIs with known accuracy and lag characteristics.

Phase 1 Completed Fixes

Fix	Description	Files Changed
Fix 1	DXY label corrected to Fed Broad USD with subtitle "Broad USD vs 25 currencies (not ICE DXY)"	App.jsx
Fix 2	VIX sourced from FRED VIXCLS (real CBOE data). Added legend for 15-20 Normal, 20-30 Elevated, 30+ Extreme	index.js, Components/VIX.js
Fix 3	Fed Funds Rate hardcoded 3.50-3.75%. June cut probability hardcoded 6% from CME FedWatch	index.js
Fix 4	QTR earnings replaced Claude estimates with real Finnhub/SEC data (AAPL \$10.15, MSFT \$4.27 verified)	Components/Quarterly.js
Fix 4b	Government reports anchored to real values: GDP +2.0%, Deficit \$160.7B, Debt \$38.9T, Trade -\$60.3B	index.js
Fix 5	Monthly TIC data replaced Claude estimates with real Treasury API (Taiwan \$1,239B, Germany \$693B, U.S. \$1,239B)	api/tic.js, index.js
Fix 6a	Added "Changes shown are vs prior month (MoM)" label to TIC Section	App.jsx
Fix 6b	Added week-ending date to Fed Balance Sheet weekly change	App.jsx
Fix 6c	Fed Balance Sheet badge now shows actual FRED data date not today's date	components.js
Fix 7	Intel briefing anchored to verified live prices instead of generic estimates	fetcher.js
Fix 8	WTI/Brent routed from FRED EIA data (accurate, 1-week lag) instead of ETF proxies	index.js
Fix 9	WTI/Brent upgraded to real-time via OilPriceAPI.com (~30-min delay) instead of fetchers.js	api/oil.js
Fix 10	Cache busting added to all fetchLive calls (no-store + timestamp) to ensure cross-platform freshness	fetcher.js
Fix 11	Auction status shows PENDING when Treasury BTC data not yet published	api/treasury.js, App.jsx
Fix 12	Auctions sorted by maturity descending: 30Y > 10Y > 7Y > 5Y > 3Y > 2Y	api/auctions.js

Fix 13	Hidden blank "High yield: %" and "avg undefined%" in auction capx	appsjsx
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Phase 1 Data Architecture

All data falls into one of four tiers:

Tier	Type	Examples	Update Frequency
1	Real-time	Gold, Silver, Mag7, TIC, Auctions	Live / same-day
2	Real with lag	WTI/Brent (OilPriceAPI ~30min), VIX/Yields (FRED 30-day), CPI/PPI (monthly)	
3	Hardcoded	Fed rate, FOMC dates, Intel price anchors, trade balance	Manual update required
4	Pending	Auction tail (true), Headline DXY, Futures curves	Requires IB or paid API

Phase 1 API Stack

API	Data Provided	Cost	Key Required
gold-api.com	Gold, Silver spot prices	Free	Yes (Vercel env)
Finnhub	Mag7, SPX/NDX/DJI, Earnings	Free tier	Yes (Vercel env)
FRED (St. Louis Fed)	Yields, CPI, PCE, GDP, Fed Balance Sheet, VIX, TGA	Free	Yes (Vercel env)
open.er-api.com	EUR/USD, USD/JPY, GBP/USD	Free	No
Treasury Fiscal Data API	Auction BTC, Indirect, Dealer, High Yield	Free	No
Treasury TIC (slt_table5.txt)	Foreign holdings of US Treasuries	Free	No
OilPriceAPI.com	WTI crude, Brent crude (near real-time)	Free tier	Yes (Vercel env)
Anthropic Claude API	Intel briefing, Fed narrative, Gov reports	Paid per token	Yes (Vercel env)

PHASE 2 — Richer Visualization & Interpretation (PLANNED)

Phase 2 Objective

Transform the dashboard from a data display into a macro interpretation tool. Add trend context, composite cycle indicators, and a reference framework so every number can be understood in historical and theoretical context.

Item	Feature	Description	Data Source	Priority
2A	Intel Live Prices	Pass real Daily tab prices into Intel Claude Existing API	eliminate all historical anchors	HIGH
2B	Trend Sparklines	12-month inline sparklines for yields, gold, FX, historical (file)	FRED, historical (file)	HIGH
2C	Dalio Deleveraging Meter	Composite 0-100 gauge scoring Beautiful FRED/Treasury data	FRED/Treasury data	HIGH
2D	Debt Cycle Stage Indicator	Horizontal progress bar: Stage 1 (Sound Money) through Stage 6 (Monetization)	Money, stage 6 (Monetization)	HIGH
2E	Macro Cheat Sheet Panel	1-1 app collapsible reference: Dalio framework, signs, rules	Dalio content, signs, rules	MEDIUM
2F	Missing Indicators	Add: 5Y5Y breakeven, corporate credit spreads, (free) interest payments, M2, Bitcoin	FRED, (free) interest payments, M2, Bitcoin	MEDIUM
2G	EODHD Integration	Real-time NDX direct, headline DXY ~98, +122/mo data for sparklines	EODHD, +122/mo data for sparklines	MEDIUM

Phase 2 Design Principles

- Mobile-first — all additions must work on iPhone PWA
- No breaking changes — each feature added independently and tested before next
- Real data first — no Claude estimates where FRED/Treasury data exists
- One change at a time — no multi-file simultaneous edits
- Verify before deploy — test each API endpoint before wiring to UI

PHASE 3 — Investment Guidance & Action Layer (FUTURE)

Phase 3 Objective

Turn MacroWatch from a monitoring tool into an actionable decision-support system. Generate investment theses based on Dalio scenario frameworks, provide positioning signals tied to real indicator thresholds, and integrate Interactive Brokers for true auction tail data and futures.

Item	Feature	Description	Dependency
3A	Scenario Framework	5 scenarios tied to Debt Cycle Stage: Muddling Through, Stagflation, Ugly Inflation, Reserve Currency Collapse, Debt Default	Phase 2 complete
3B	Dynamic Thesis Generator	Rule-based positioning signals: "VIX>30 + portfolio underperforming = Defensive"	Phase 2 complete
3C	Wealth Protection Playbook	Static but updated framework: core holdings, hedges, avoid/reduce per Dalio late-cycle scenarios	Phase 2 complete
3D	IB API Integration	True auction tail (when-issued yield), futures (Basis, USDT, etc), USD/SGD	Phase 2 complete
3E	Alert System	Threshold breach notifications: auction tail > 2bps, TGA < \$200B, VIX > 30, etc	Phase 2 complete
3F	Portfolio Simulator	Hypothetical allocations backtested against historical Dalio cycle periods	Phase 2 complete

Phase 3 Alert Thresholds

Alert	Threshold	Severity	Dalio Significance
Auction Tail widens	> 2bp	Warning	Demand shock — buyers wanted higher yield than market expected
Indirect Bidders drop	< 55% for 2 consecutive	Critical	Foreign demand retreat — reserve diversification accelerating
Dealer Takedown surges	> 20%	Critical	No one else wanted it — forced buying, yield pressure ahead
VIX spikes	> 30	Warning	Fear entering market — late-cycle risk decompressing
Gold breaks \$5,000	> \$5,000	Watch	Reserve currency confidence loss accelerating
TGA drops sharply	< \$200B	Warning	Treasury liquidity stress — issuance surge incoming
10Y yield crosses	> 5%	Warning	Debt service costs escalating — sustainability risk
Fed Funds Futures	> 75bps cuts H2	Watch	Market pricing policy capitulation on inflation